

Course 2 glossary

Terms and definitions from Course 2

A

Application programming interface (API): A protocol that enables one application to interact with another application

Architecture: A framework that defines the design of a technical solution

Array: A collection of values of the same data type

B

Batch processing: A method of collecting large volumes of data over a period of time, then processing it all at once

Boolean: A data type with only two possible values, usually true or false

C

Classification: The process of grouping items into a predefined category based on common characteristics

Cloud data storage: A solution that enables organizations to keep, access, and maintain digital data in off-site, cloud-based servers

Clustered column: A user-defined table property that arranges storage blocks based on the values within their columns

Clustered table: A table in which column order is defined by the user, using clustered columns

Clustering: Grouping data points that are similar to each other in some way

Comma-Separated Values (CSV) file: A delimited text file that uses a comma to separate field values

Context: The condition in which something exists or happens

Credentials: A username and password

D

Data catalog: A centralized inventory of an organization's data assets

Data connection: A link formed between a data source and its storage location

Data curation: The process of selecting and organizing data to ensure it's useful and of high quality

Data discovery: The process of finding and understanding data assets within a dataset or data ecosystem, and identifying relevant patterns and relationships

Data governance: A process for ensuring the formal management of a company's data

Data ingestion: The process of obtaining, importing, and processing data for later use or storage

Data lake: A database system that stores large amounts of raw data in its original format until it's needed

Data lakehouse: A hybrid data architecture that combines the features of a data lake with those of a data warehouse

Data mart: A subject-oriented database that can be a subset of a larger data warehouse

Data model: A concept for organizing data elements and how they relate to one another

Data pipeline: A series of processes that transport data from different sources to a destination for storage and analysis

Data source: The place where data was generated

Data unification: The process of integrating data from diverse sources to create a consolidated view

Data warehouse: A database that consolidates data from multiple source systems for data consistency, accuracy, and efficient access

Database partitioning: The process of dividing data into separate data segments or partitions that can be managed and accessed separately

Database schema: A way of describing how a database is organized

Dataproc: A fully managed service that maximizes open-source data tools for batch processing, querying, streaming, and machine learning

Denormalized data: Stores repeated information in one or more tables

F

Float: A number that contains a decimal

Foreign key: A column within a table that is a primary key in another table

Functional partitioning: A process for dividing data into separate segments, in which data is grouped based on how it's used in different parts of the database system

G

Google Colab: A cloud-hosted version of Jupyter Notebook that allows users to write and execute Python in a browser, with no configuration required

H

Hadoop: A framework that distributes the processing of data across clusters of computers

Horizontal partitioning: A process for dividing data into separate segments, in which each partition or shard is its own data storage, but all shards follow the same organizational layout

Hyperparameters: Parameters whose value is used to control the learning process

I

Input data: Any information that a user sends to a computer

Integer: A data type that contains whole numbers

J

JavaScript Object Notation (JSON): A text-based data-interchange format

L

Linear regression: A technique that estimates the linear relationship between a continuous dependent variable, and one or more independent variables

M

Machine learning: The use and development of algorithms and statistical models to teach computer systems to analyze and discover patterns in data

Master data management (MDM): A discipline and framework for managing and maintaining critical data assets in order to achieve a single, consistent view of data across the organization

Metadata: Data about data

N

Nested data structure: A structure that organizes data within other data structures, forming a hierarchy of information

Normalized data: Organizes related fields into different tables, and maintains defined relationships between columns in these different tables

O

Output data: Any information that a computer sends to a user

P

Partitioned table: A large table that is divided into smaller segments, or partitions, to improve query performance, and control costs by reducing the amount of data a query needs to process

Partition pruning: The process of eliminating unnecessary or irrelevant data from consideration when running a query

Primary key: An identifier that references a column in which each value is unique

PySpark: A language for creating more automated analyses and pipelines

R

Ranking: The process of using characteristics to order items by likelihood

Relational database: A database that contains a series of tables that can be connected to form relationships

S

Schema: A way of describing how something, like data, is organized

Semi-structured data: Data that has some structure but is not as precisely organized as structured data

Software development kit (SDK): A set of software-building tools that includes groups of code libraries

SQL dialect: A version of SQL that is unique to a specific database

Streaming data processing: A method of processing data as it's received

String: A sequence of characters and punctuation that contains textual information

Struct: A collection of named fields

Structured data: Data organized in a certain format, like rows and columns

Syntax: The predetermined structure of a language that includes all required words, symbols, and punctuation, and their proper placement

System of record: A data-storage system that serves as the source of truth for an organization's data related to processes or systems

T

Transactional database: A data-storage system that maintains individual rows of data, rather than columns

U

Unstructured data: Data that is not organized in any easily identifiable way

V

Vertical partitioning: A process for dividing data into separate segments, in which each partition maintains a unique piece of the fields for items in the data store